

Bouira University

Cloud Computing
Part 1

Master: GSI & ISIL

Department of Computer Science.

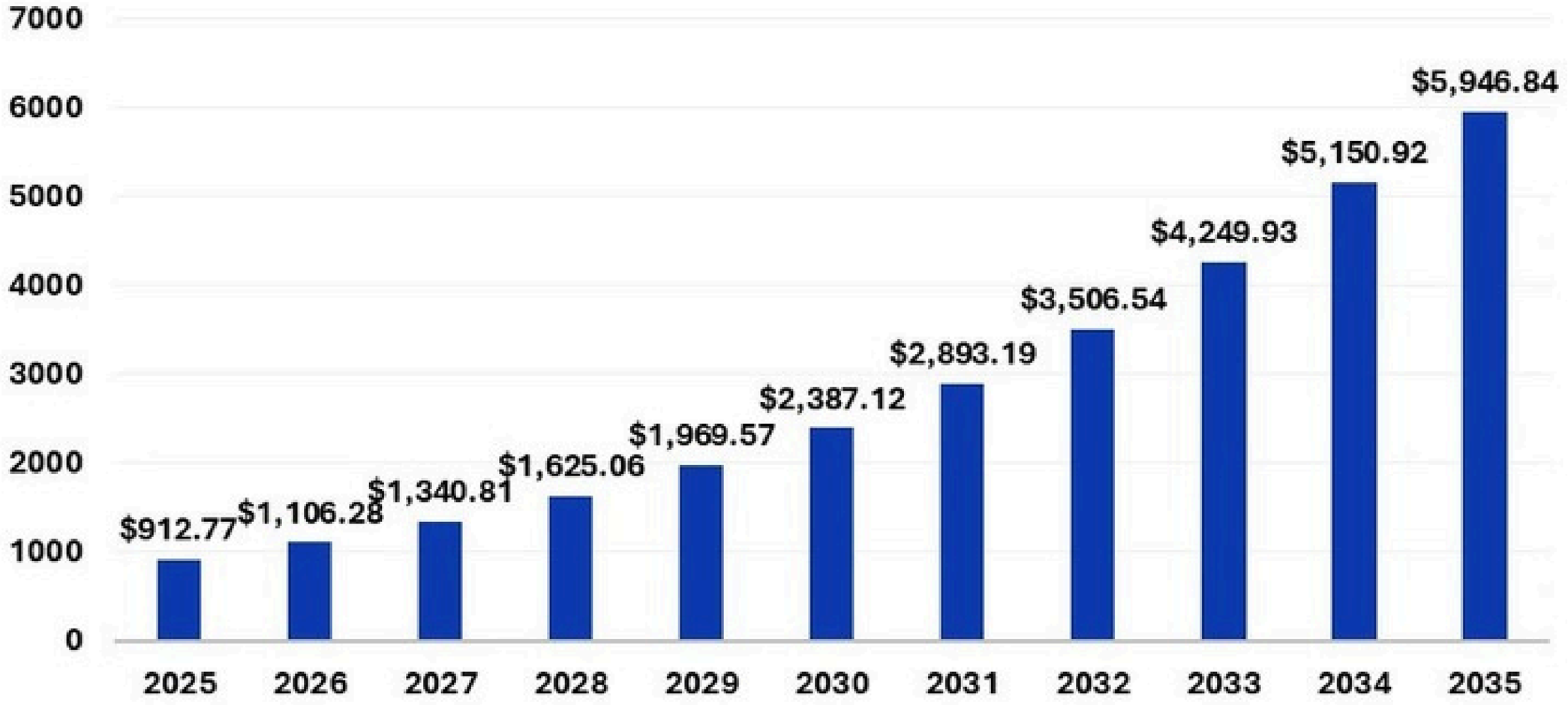
Presented by
Dr S.BENBATATA

Introduction to Cloud Cmputing

Cloud computing is one of the most important technologies today and will remain a critical paradigm in the coming years. It has made significant inroads across all sectors, including scientific research, e-commerce, the IT industry, deep learning, the Internet of Things (IoT), healthcare applications, and more

Precedence RESEARCH

Cloud Computing Market Size 2025 to 2035 (USD Billion)



Source: https://www.precedenceresearch.com/cloud-computing-market

Introduction to Cloud Cmputing

Cloud computing enables a wide range of applications and projects, from providing infrastructure to software as a service.



Introduction to Cloud Cmputing

Key Concepts

What do you mean by **Distributed Systems?**



Introduction to Cloud Cmputing

Key Concepts

01

Distributed Systems

A distributed system is a set of independent computers and devices that work together and appear to users as a single system

Introduction to Cloud Cmputing

Key Concepts

What do you mean by **virtualization?**



Introduction to Cloud Cmputing

Key Concepts

02 **Virtualization**

Virtualization is a core technology in Cloud Computing that abstracts software and physical computing resources such as memory, storage, and network, etc., by installing specific software that enables this abstraction.

It create virtual machines from these resources, allowing multiple operating systems and applications to run simultaneously on a single physical machine.

Introduction to Cloud Cmputing

Key Concepts

Runtime Environment



Introduction to Cloud Cmputing

Key Concepts

03

Runtime Environment

that simulates the conditions required for applications to run in the cloud, including the operating system, libraries, and any additional tools required for running applications.

Introduction to Cloud Cmputing

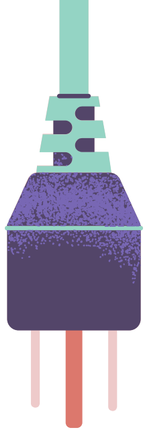
Key Concepts

04

Multi-tenancy

Multi-tenancy in cloud computing means that a single instance of software or hardware is used by multiple users, known as tenants. Each user has its own environment, data, and usage. These resources are logically isolated from others, even though they share the same physical resources

What is Cloud

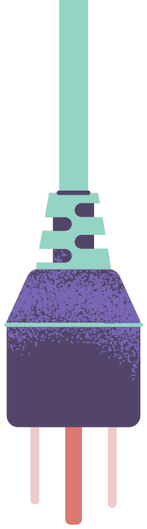


is a technology that provides access to IT systems and services over the internet. Typically, cloud computing provides different types of services, such as storage, networking, data and programs, where users can access the data from anywhere at any time just by using a simple device.

There are many definitions for cloud computing, such as the one provided by the National Institute of Standards and Technology (NIST).



What is Cloud



The NIST Definition of Cloud Computing:

Cloud computing is a model for enabling ubiquitous, on-demand network access to a shared pool of configurable computing resources.

This cloud model is composed of five essential characteristics, three service models, and four deployment models



https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf

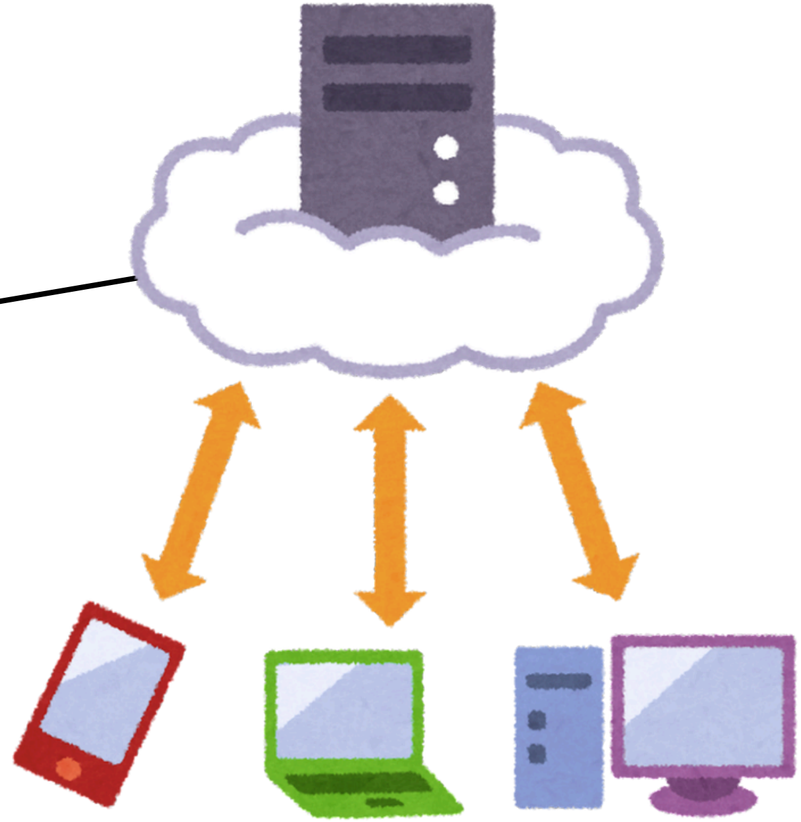
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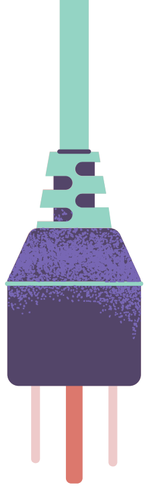




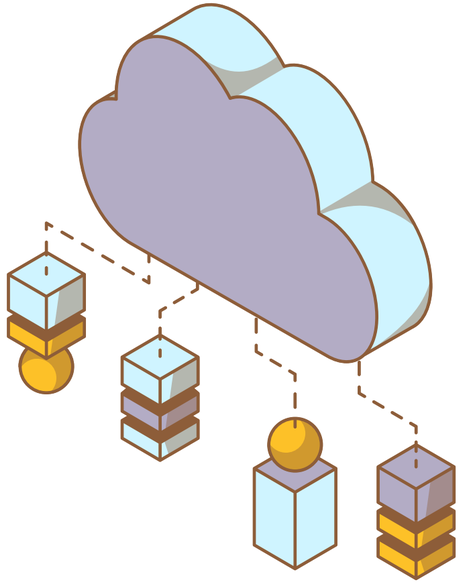


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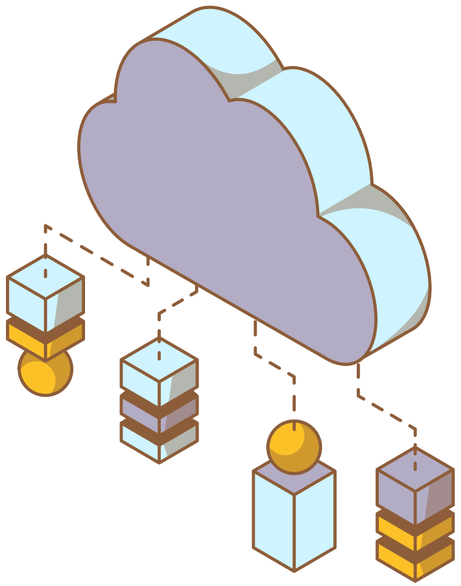
01 Google Cloud Platform



Some apps that are based on cloud computing:;

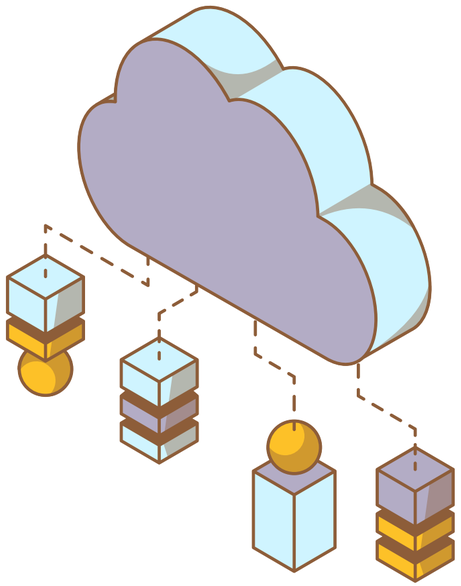


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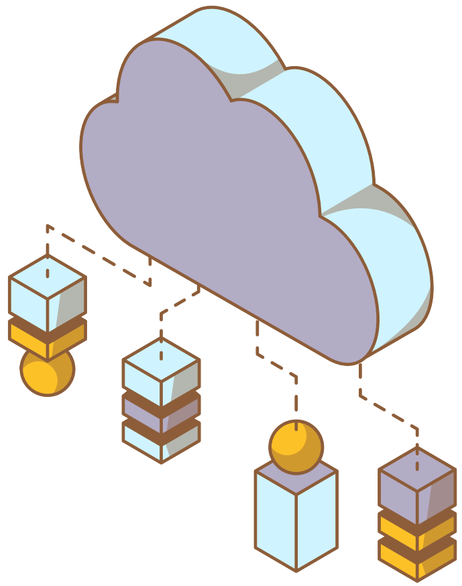
02 Google Drive and GitHub

Introduction



03 Netflix

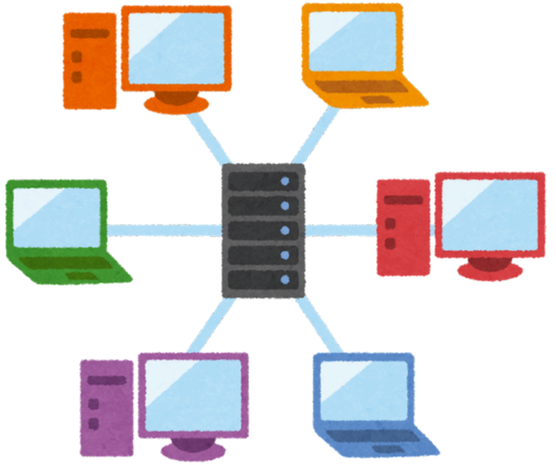
Introduction



04

Adobe Creative Cloud such Photoshop, Illustrator ect.

Introduction

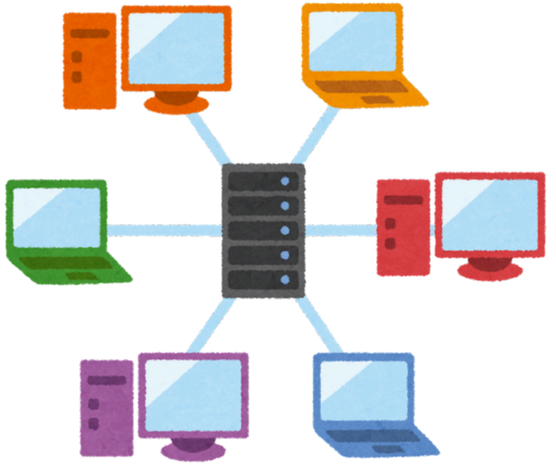


History

Before cloud computing technology emerged, there were client/server computing models, where data and applications were hosted on centralized servers(on-premises). This technology provided services through a single provider, which made it easier, but faced many challenges, such as :

- physical distance from users, which impacts capacity and performance.
- The increasing volume of requests slows down the analysis
- process, resulting in server overload.
- High energy consumption,
- Security problem .

Introduction



History

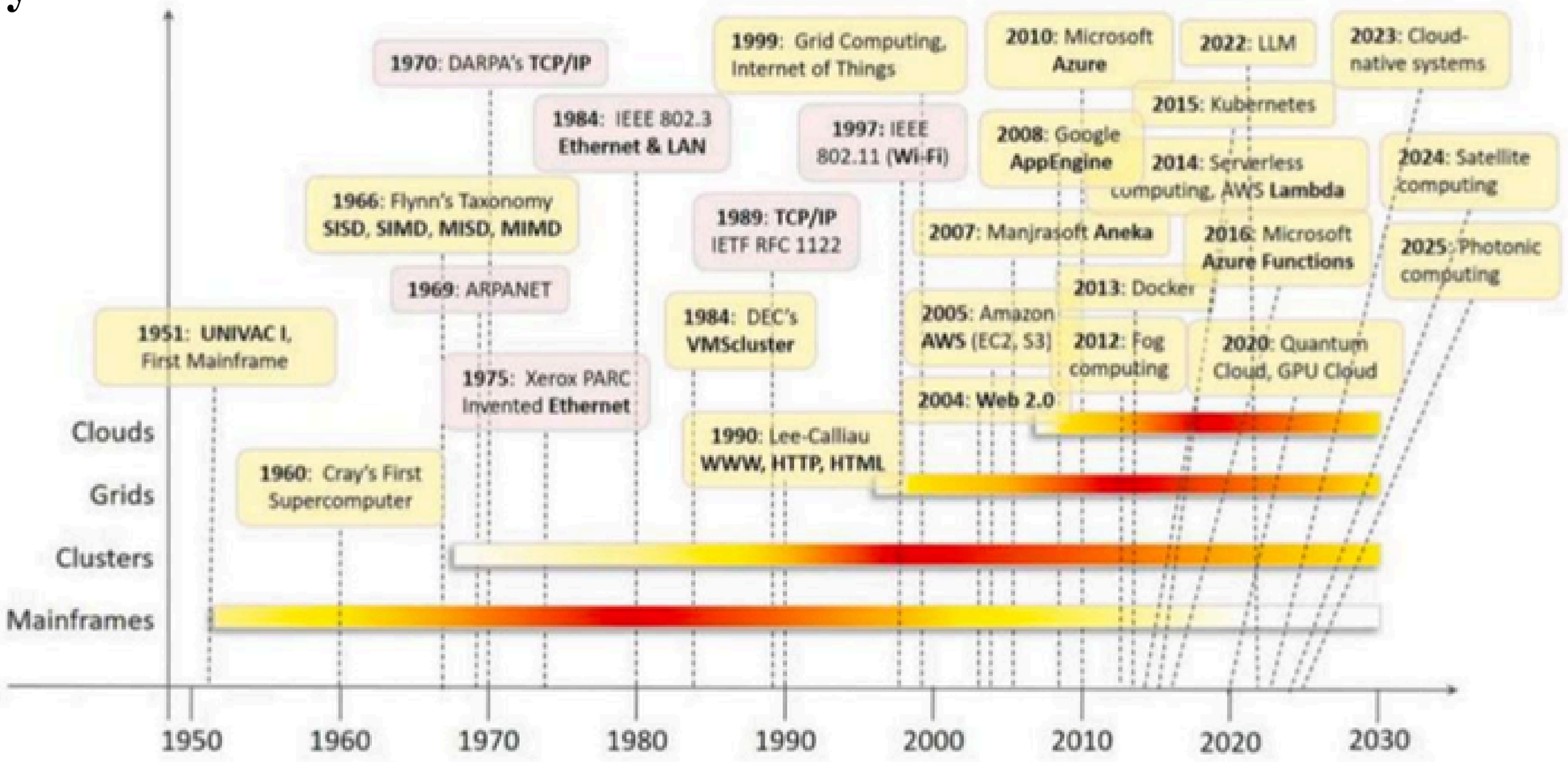
The concept of cloud computing as a utility was first proposed by John McCarthy in 1961, but the technology did not exist at that time.

The modern cloud era began in 1999 with Salesforce.com as the first company to offer cloud computing services, followed by Amazon Web Services with its core infrastructure services in 2006, and later Google App Engine and Microsoft Azure between 2008 and 2010.

Besides these, other major companies like HP, Oracle, and Alibaba Cloud entered the market.

Introduction

History



Advantages of Cloud Computing



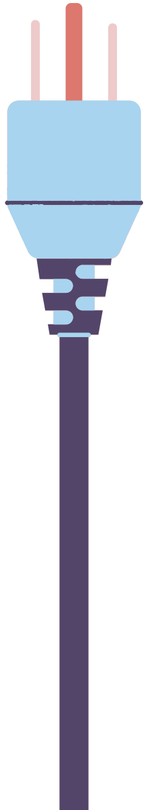
Trade Capital Expense (CapEx) for Variable Expense (OpEx)

CapEx (Capital Expense):



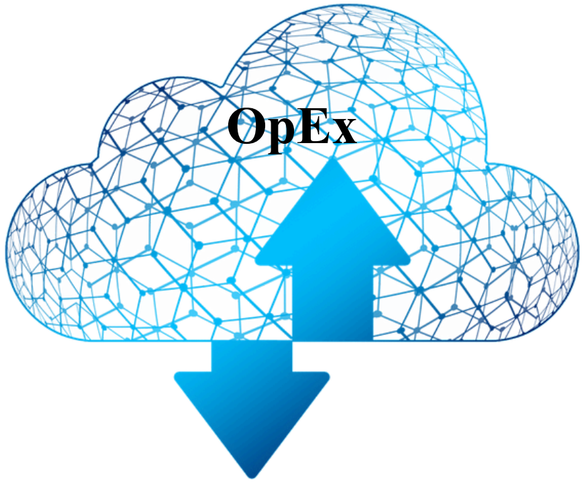
OpEx (Operational Expense)





Advantages of Cloud Computing

Trade Capital Expense (CapEx) for Variable Expense (OpEx)



CapEx

CapEx (Capital Expense):



OpEx (Operational Expense)

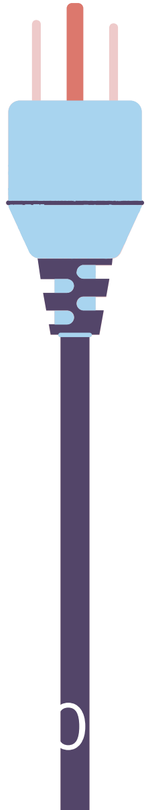


Advantages of Cloud Computing



Trade Capital Expense (CapEx) for Variable Expense (OpEx)

cloud computing is based on a simple paradigm: keeping capital costs (CapEx) low and maximizing operational expenditure (OpEx). This means users only pay for what they actually use. For businesses, this approach provides many benefits, such as flexibility and cost savings. Instead of buying expensive hardware and managing everything themselves, they can simply use cloud services and let the provider handle the technical side of the infrastructure, such as installation, updates, security, problems, and maintenance.



Advantages of Cloud Computing

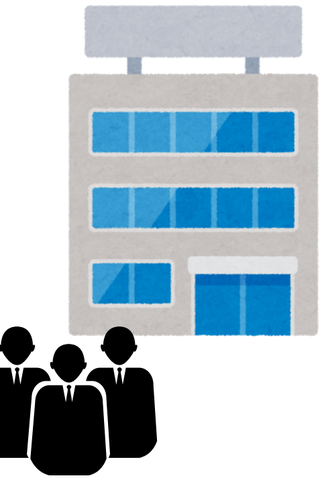
Benefit from Massive Economies of Scale

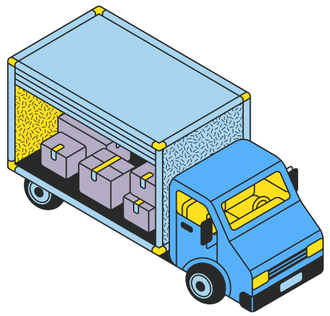


AWS

Google









Advantages of Cloud Computing

Benefit from Massive Economies of Scale

what is the benefit of that?

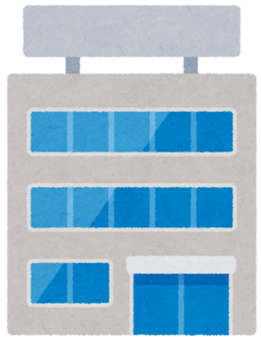


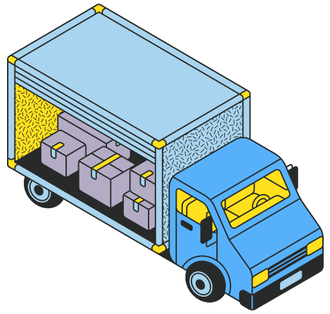
Advantages of Cloud Computing

AWS

Google

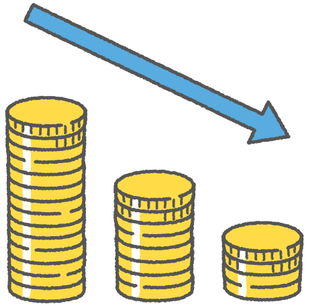
Benefit from Massive Economies of Scale







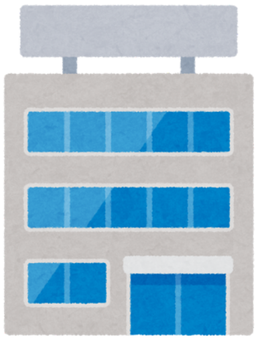




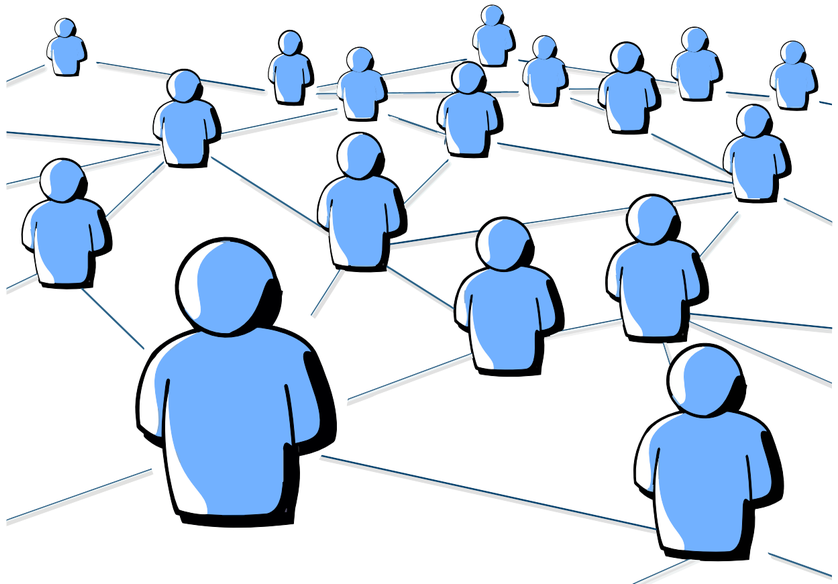
Advantages of Cloud Computing



Benefit from Massive Economies of Scale









Advantages of Cloud Computing

Stop Guessing Capacity

If a business runs an e-commerce store on local servers, they have to predict human traffic.





predict human traffic,
During peak seasons, ect.



Advantages of Cloud Computing

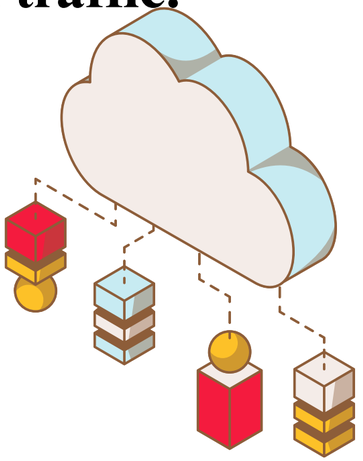
Stop Guessing Capacity

If a business runs an e-commerce store on local servers, they have to predict human traffic.



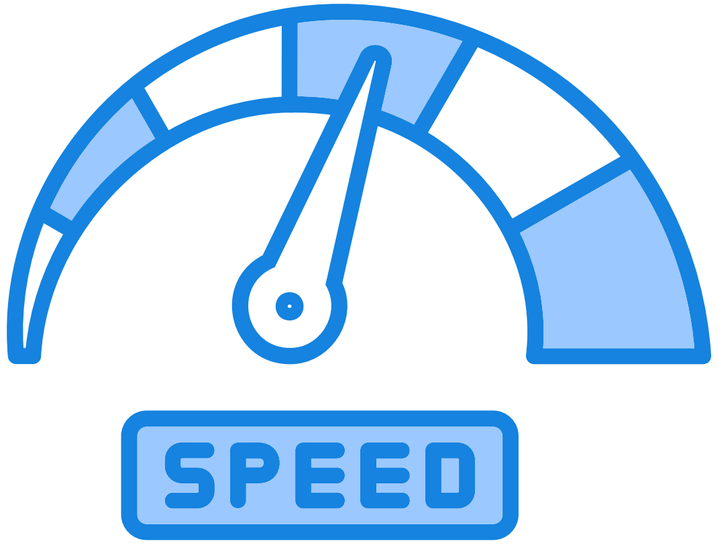
Over-provisioning

Under-provisioning



Advantages of Cloud Computing

Increase Speed



With cloud computing, new IT resources are just a click away, companies no longer need to waste time and money making new resources available for developers, and can instead focus on product development and delivery.

Advantages of Cloud Computing

Stop Spending Money Running and Maintaining Data Centers



Instead of wasting time managing and maintaining costly and complex on-premises data centers, cloud computing lets you focus on your projects

Advantages of Cloud Computing

Go Global in Minutes

Cloud computing allows you to deploy solutions on a global scale across multiple regions with low latency and low cost in just a few clicks and less time.

6 advantages of cloud computing

